

# Pharma Agentic AI – Analysis Report

Automated opportunity assessment powered by Agentic AI (Master + Worker agents).

|                  |                                                                                                                                                           |              |                                              |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------|
| Query            | Perform a complete opportunity assessment for MOLX in the Indian Asthma market including market, EXIM, patents, trials, internal insights and guidelines. |              |                                              |
| Molecule         | MOLX                                                                                                                                                      | Therapy area | Asthma                                       |
| Region / Country | India                                                                                                                                                     | Agents run   | MARKET EXIM<br>PATENT TRIALS<br>INTERNAL WEB |
| Generated on     | 08 Dec 2025, 11:25 PM                                                                                                                                     |              |                                              |

## Executive Summary

- Market & growth view: Total market size: 6722.8 Mn USD with average 5Y CAGR 9.3%. Records: 16.
- Supply and trade dynamics: Total imports: 27940.6 tonnes, total exports: 27449.0 tonnes across 272 records.
- IP barriers and freedom-to-operate: Found 158 relevant patents. Earliest expiry year: 2003.
- Clinical development and pipeline: Found 13 clinical trials matching the criteria.
- Internal strategy and unmet needs: MOLX Opportunity Assessment in Indian Asthma Market
- Guidelines and real-world evidence: ASTHMA 2024 guidelines recommend step-wise treatment based on severity and patient phenotype, while real-world evidence highlights suboptimal adherence due to access, affordability, and health literacy barriers in emerging markets, with regional variation in outcomes.

## Agent-level Insights

|                                                                          |
|--------------------------------------------------------------------------|
| <b>IQVIA Insights Agent</b>                                              |
| <b>High-level summary</b>                                                |
| Total market size: 6722.8 Mn USD with average 5Y CAGR 9.3%. Records: 16. |

Structured details

Table:

- **molecule:** MOLX **indication:** Asthma **country:** India **year:** 2018 **sales\_usd\_mn:** 133.55 **cagr\_5y:** -2.97 **competitors:** 6 **patient\_burden\_index:** 0.48
- **molecule:** MOLX **indication:** Asthma **country:** India **year:** 2017 **sales\_usd\_mn:** 497.55 **cagr\_5y:** 10.73 **competitors:** 20 **patient\_burden\_index:** 0.63
- **molecule:** MOLX **indication:** Asthma **country:** India **year:** 2015 **sales\_usd\_mn:** 849.71 **cagr\_5y:** 14.35 **competitors:** 5 **patient\_burden\_index:** 0.36
- **molecule:** MOLX **indication:** Asthma **country:** India **year:** 2017 **sales\_usd\_mn:** 603.09 **cagr\_5y:** 11.43 **competitors:** 20 **patient\_burden\_index:** 0.26
- **molecule:** MOLX **indication:** Asthma **country:** India **year:** 2024 **sales\_usd\_mn:** 525.85 **cagr\_5y:** 8.78 **competitors:** 21 **patient\_burden\_index:** 0.55
- ... (5 more records)

EXIM Trade Agent

High-level summary

Total imports: 27940.6 tonnes, total exports: 27449.0 tonnes across 272 records.

Structured details

Table:

- **molecule:** MOLX **formulation:** Tablet **country:** India **year:** 2015 **import\_tonnes:** 159.88 **export\_tonnes:** 97.59
- **molecule:** MOLX **formulation:** Capsule **country:** India **year:** 2024 **import\_tonnes:** 167.83 **export\_tonnes:** 144.32
- **molecule:** MOLX **formulation:** Injection **country:** India **year:** 2019 **import\_tonnes:** 50.73 **export\_tonnes:** 26.51

- **molecule:** MOLX **formulation:** Suspension **country:** India **year:** 2018 **import\_tonnes:** 161.52 **export\_tonnes:** 19.47
- **molecule:** MOLX **formulation:** Injection **country:** India **year:** 2018 **import\_tonnes:** 156.3 **export\_tonnes:** 169.86
- ... (5 more records)

## Patent Landscape Agent

### High-level summary

Found 158 relevant patents. Earliest expiry year: 2003.

### Structured details

#### Patents:

- **patent\_id:** US1002042 **molecule:** MOLX **indication:** Asthma **assignee:** Sunrise Therapeutics **filing\_year:** 2022 **expiry\_year:** 2039 **status:** Pending **fto\_flag:** Medium
- **patent\_id:** US1002677 **molecule:** MOLX **indication:** Asthma **assignee:** OncoGenix **filing\_year:** 2009 **expiry\_year:** 2019 **status:** Expired **fto\_flag:** Medium
- **patent\_id:** US1003343 **molecule:** MOLX **indication:** Asthma **assignee:** NovaCure **filing\_year:** 2015 **expiry\_year:** 2036 **status:** Expired **fto\_flag:** Low
- **patent\_id:** US1003913 **molecule:** MOLX **indication:** Asthma **assignee:** PrimeCare **filing\_year:** 2006 **expiry\_year:** 2026 **status:** Granted **fto\_flag:** High
- **patent\_id:** US1004134 **molecule:** MOLX **indication:** Asthma **assignee:** MedLife Corp **filing\_year:** 2013 **expiry\_year:** 2026 **status:** Granted **fto\_flag:** High
- ... (5 more records)

## Clinical Trials Agent

### High-level summary

Found 13 clinical trials matching the criteria.

Structured details

Table:

- **trial\_id:** CT-009580 **molecule:** MOLX **indication:** Asthma **phase:** Phase III **status:** Active  
**sponsor:** ABC Pharma **country:** India
- **trial\_id:** CT-009652 **molecule:** MOLX **indication:** Asthma **phase:** Phase II **status:** Recruiting  
**sponsor:** NovaCure **country:** India
- **trial\_id:** CT-019510 **molecule:** MOLX **indication:** Asthma **phase:** Phase II **status:** Active  
**sponsor:** OncoGenix **country:** India
- **trial\_id:** CT-026574 **molecule:** MOLX **indication:** Asthma **phase:** Phase IV **status:** Completed  
**sponsor:** NovaCure **country:** India
- **trial\_id:** CT-027216 **molecule:** MOLX **indication:** Asthma **phase:** Phase II **status:** Terminated  
**sponsor:** GlobalBio **country:** India
- ... (5 more records)

Internal Knowledge Agent

High-level summary

MOLX Opportunity Assessment in Indian Asthma Market

Structured details

Documents:

- asthma\_strategy\_2024.pdf
- respiratory\_field\_insights.pdf

Key Opportunities:

- Once-daily inhaler for better adherence and convenience
- Expansion into pediatric asthma segment with low-dose options
- Targeting elderly population with safer, steroid-free formulations
- Alternative therapies for steroid-intolerant patients
- Low-cost fixed-dose combinations for Tier-2/3 cities

**Key Risks:**

- Exclusivity constraints due to patent expiration and generic competition
- Evidence gaps in pediatric use of MOLX
- High operational costs for targeted low-dose formulations
- Risk of low adherence due to inhaler technique issues
- Affordability and pricing challenges in Tier-2/3 cities

**Web Intelligence Agent**

**High-level summary**

ASTHMA 2024 guidelines recommend step-wise treatment based on severity and patient phenotype, while real-world evidence highlights suboptimal adherence due to access, affordability, and health literacy barriers in emerging markets, with regional variation in outcomes.

**Structured details**

**Links:**

- **title:** Asthma guideline (synthetic) **url:** <https://example-guideline.com>
- **title:** Asthma RWE summary **url:** <https://example-rwe.com>

**Key Updates:**

- Combination therapy is recommended when monotherapy fails to achieve asthma control.
- Non-pharmacological measures, such as lifestyle, diet, and exercise, are core components of asthma management.
- Suboptimal adherence to guideline-directed therapy is observed in low-resource settings, driven by access, affordability, and health literacy challenges.

This report is generated by the Pharma Agentic AI prototype for EY Hackathon 6.0. Insights are based on simulated / mock data sources and should not be used as a substitute for validated market research.